

ToolCall M13 scaling study

Fixed 12.9M-parameter model · comparison across training-token budgets

COMPLETED

3/3 runs

BEST GENERAL LOSS

3.0837 · High · 460M

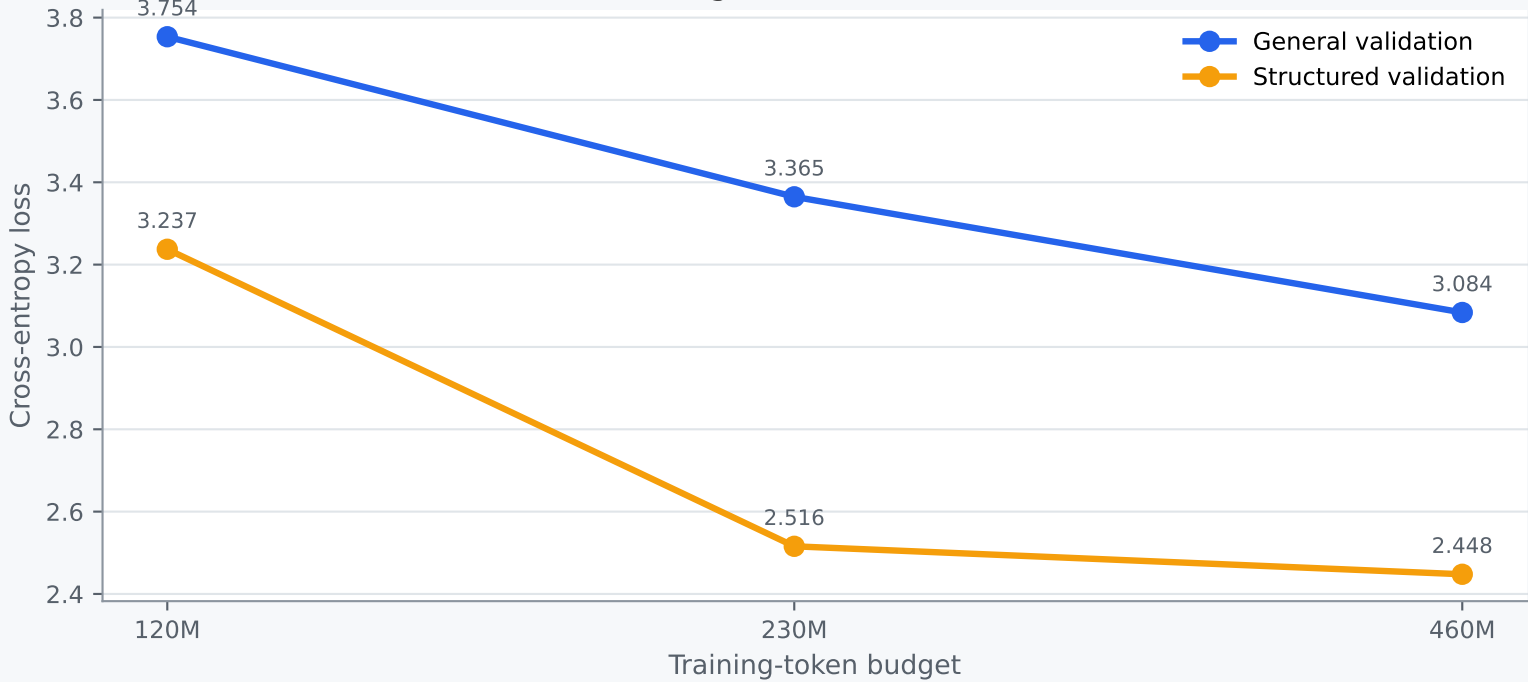
BEST STRUCTURED LOSS

2.4478 · High · 460M

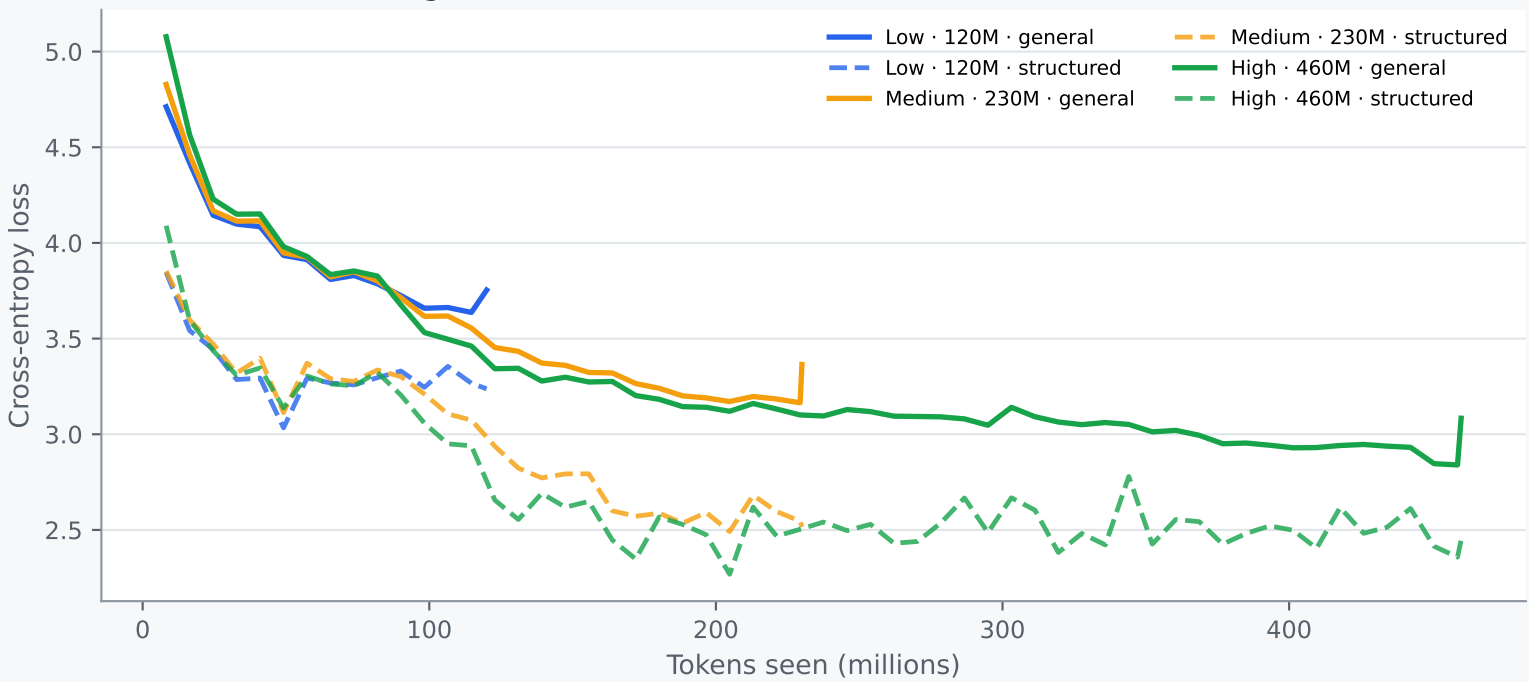
TOTAL GPU TIME

3.29 hours

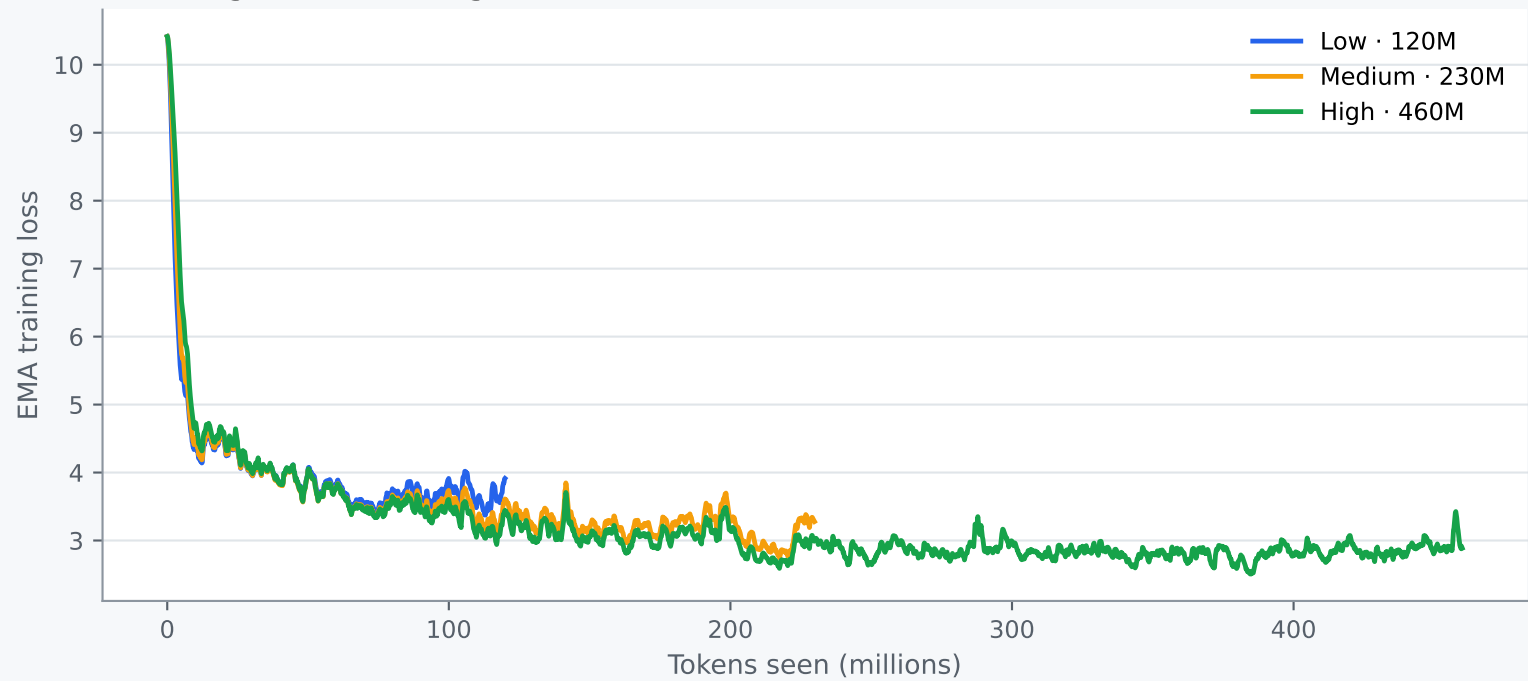
Final validation loss vs token budget



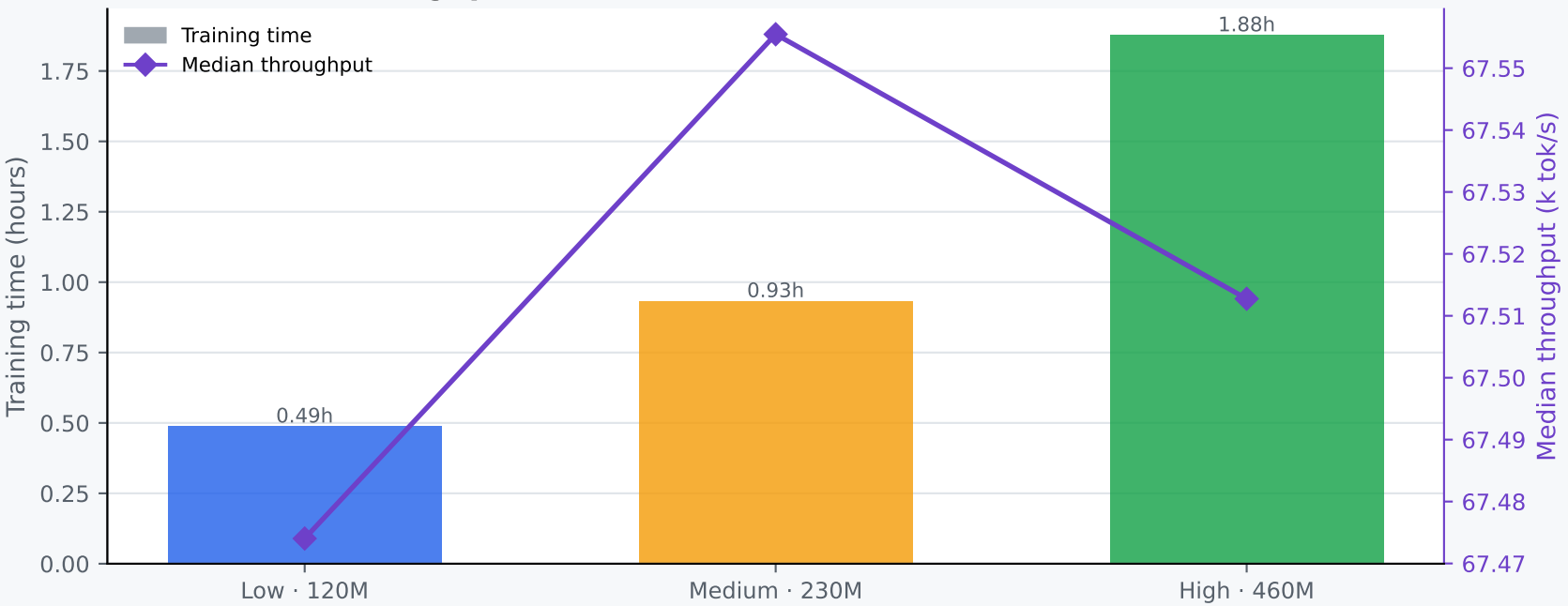
Validation learning curves



Training-loss convergence



Runtime and throughput



Run comparison

Run	Status	Tokens	General loss	Structured loss	Train time	Median speed	Peak VRAM
Low · 120M	completed	120.0M	3.7535	3.2373	0.49 h	67.5k tok/s	2.40 GiB
Medium · 230M	completed	230.0M	3.3647	2.5157	0.93 h	67.6k tok/s	2.40 GiB
High · 460M	completed	460.0M	3.0837	2.4478	1.88 h	67.5k tok/s	2.40 GiB